COGNITIVE FUNCTION & NEUROPROTECTION

# Selank — the full review

TUFTSIN-DERIVED SYNTHETIC PEPTIDE

*This is the deeper look behind the short version — what the Russian anxiety trials actually show, how that evidence compares to FDA-standard trials, common misunderstandings, and what your consultation covers.*

## What it is

Selank is a synthetic seven-amino-acid peptide engineered in Russia in the 1990s as a chemically stabilized analog of tuftsin, a naturally occurring fragment of the antibody protein IgG involved in normal immune signaling. It was designed to resist the enzymes that break down natural tuftsin quickly, extending its activity in the body. It is most often used as an intranasal spray for calm and stress resilience, marketed as a non-sedating alternative to a benzodiazepine, though gray-market vendors also sell injectable versions.

Selank is a licensed prescription medicine in Russia, developed and studied through Russian government-affiliated pharmacology institutes, though it has never been reviewed by the FDA and carries no approval, in any form, in the United States. It sits in FDA's "nominated but withdrawn" compounding-substance table, where FDA states it lacks important information regarding safety issues raised by Selank administered to humans.

The human research behind it is real but modest — a handful of small Russian trials, developed and tested almost entirely by the same government-affiliated institutions that created the compound, published mostly in Russian-language journals, without independent replication in the West.

## What people use it for

| USED FOR | EVIDENCE | WHAT THAT MEANS |
|---|---|---|
| **Anxiety, including generalized anxiety** | **Human trials** | Small Russian trials report relief over 2-4 weeks, with no outside replication. |
| **Stress resilience** | **Limited human data** | Overlapping data from the same trial programs touches on stress symptoms, but no dedicated trial isolates this as a primary endpoint. |
| **Cognitive focus** | **Animal data only** | Marketed as a nootropic, but the supporting evidence for BDNF regulation and memory protection is rat-only. |
| **Immune modulation** | **Animal data only** | As a tuftsin analog, documented cytokine and antiviral effects exist in mice; no human immunology trials have been published. |
| **Addiction and withdrawal support** | **Animal data only** | Reduced withdrawal-related behaviors in rat and mouse studies; no published human trial in substance-use disorder, craving, or withdrawal exists. |

| USED FOR | EVIDENCE | WHAT THAT MEANS |
|---|---|---|
| **Depression** | **Theoretical** | Occasionally marketed on the strength of its BDNF and serotonergic hypotheses, but no dedicated human depression trial exists. |

## How it works

Selank's proposed mechanisms are derived almost entirely from animal and cell studies, not human mechanistic research. The longest-standing claim is that it inhibits enzymes that break down the body's own opioid-like enkephalins, prolonging their activity, proposed as distinct from how a benzodiazepine binds its receptor. In-vitro and rodent studies also report Selank alters GABA-related gene expression and interacts with serotonin signaling, and rat studies report it regulates brain-derived neurotrophic factor (BDNF) in regions tied to memory and stress. As a tuftsin analog, mouse studies also report immune and antiviral effects.

How well-established this is matters for setting expectations: not very, by the standard patients should expect elsewhere in this catalog. There is no human-confirmed mechanism — these hypotheses come from a small, overlapping set of Russian labs, have not been confirmed with human neuroimaging or receptor studies, and are at points in tension with each other. This should be described as a proposed, animal-and-cell-study-supported story, not settled pharmacology.

## What the research actually looked at

Total human exposure is small and narrow. Core trials are Russian studies of generalized anxiety and related conditions, typically a few dozen patients each — one frequently cited trial enrolled 20 patients over about two weeks. Cumulative published human subjects across all Selank trials appears to be in the low hundreds at most, nearly all recruited in Russia, with follow-up almost always days to a few weeks.

Virtually all research traces to a tight cluster of Russian, government-affiliated institutions that both developed the compound and ran the trials evaluating it — a structural independence concern. Human trials are generally small, open-label or loosely controlled, some comparing Selank to an active comparator rather than placebo, published mainly in Russian-language journals with details not consistently auditable by outside reviewers.

No independent Western replication has been identified. The broader English-language literature shows essentially no non-Russian trials of Selank, and no FDA, EMA, or other major regulator has conducted a clinical review of its efficacy data.

## Common misunderstandings

- **“It's approved and used safely in Russia, so it's proven.”** Selank is approved only in Russia, by Russia's regulator, based on trials run almost entirely by the same government-affiliated institutions that developed it, published mostly in Russian, without independent outside replication — a fundamentally different bar than FDA or EMA approval.
- **“It's non-addictive with no withdrawal, unlike benzodiazepines.”** Overstated. It's true Selank doesn't bind the same receptor complex as a benzodiazepine, and no formal physical-dependence syndrome has been documented — but that rests on short trials of days to a few weeks, with no long-term human discontinuation data.
- **“It's basically a benzodiazepine alternative with no downsides.”** Its mechanism isn't well-established, its human safety database is thin — FDA itself says it lacks important human safety information — and because it's unapproved and unregulated in the US, purity and dosing accuracy aren't guaranteed.
- **“Because it's a natural peptide fragment, it's inherently gentle and safe.”** Being structurally related to a naturally occurring immune peptide doesn't exempt a synthetic,

chemically modified analog from needing its own safety evaluation; FDA specifically flags immunogenicity and peptide-related impurity risk for compounded Selank.

### Who it is not for

- Anyone seeking a substitute for established anxiety treatment or psychotherapy.
- Pregnancy or breastfeeding, or trying to conceive.
- Children and adolescents — no pediatric safety or efficacy data exists.
- Anyone with a known or suspected autoimmune condition, given its immune-signaling mechanism and FDA's flagged immunogenicity concern.
- Anyone relying on it as a substance-use-disorder treatment — no human evidence supports this use.
- Competitive athletes subject to WADA or USADA testing.

**Competitive athletes:** Selank is not individually named on the WADA Prohibited List, but it is reasonably treated as falling under WADA's S0 ("Non-Approved Substances") category — prohibited at all times — because its only marketing approval is in Russia, not a regulator recognized for this purpose by most WADA-signatory countries. Athletes should check Global DRO or contact USADA directly and assume prohibited status by default rather than assuming permissibility from an absent name-specific listing.

### Who makes this, and under what rules

**Everything dispensed here is prepared by a licensed 503A compounding pharmacy** — not bought from a supplier and repackaged, and not sourced from an online seller.

That designation carries specific, enforceable obligations:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **Sterile preparation to a published standard.** USP standards.
- **A paper trail.** A lot number and beyond-use date on every preparation.

**A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.**

## Where this stands

**Approved in Russia, never reviewed by FDA.** A licensed anxiety medicine there, developed and tested almost entirely by the same government-affiliated institutions, without independent Western replication.

**The best human evidence supports anxiety relief specifically.** The cognitive-enhancement and stress-resilience marketing outruns what's actually been tested, which rests on animal data.

**Approved somewhere is not the same as proven by the standard this practice otherwise holds itself to.**

### Clinical detail

Selank is a stabilized synthetic analog of tuftsin, a naturally occurring immune-signaling fragment of the antibody IgG, developed at Russia's Institute of Molecular Genetics.

It is licensed by Russia's regulator, Roszdravnadzor, as a prescription intranasal anxiolytic and studied mainly through the V.V. Zakusov Research Institute of Pharmacology.

FDA's compounding-substance review flags a specific concern: potential immunogenicity and peptide-related impurities for compounded Selank, alongside a lack of human safety information.

No formal physical-dependence syndrome has been documented in the short, days-to-weeks trials available, though this has not been studied systematically over longer use.

The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment.

Selank is not an FDA-approved medicine for the uses described here. A compounded preparation is not an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality.

Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any purpose.

Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.